Lecture 28

- HW-9 due coming Friday 11pm
- HW 7-10 will be on midterm-III
- UT Sage
- Midterm-III make-up time poll
- Midterm-III practice problems

- DGCX47

Goals:

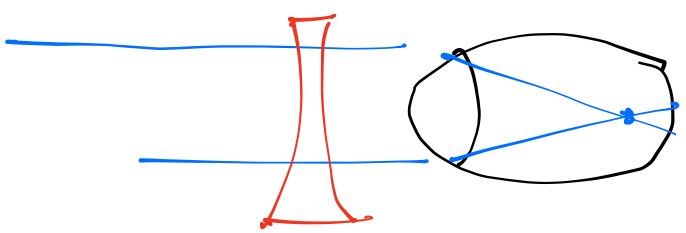
1. Lens maker's formula
2. Interference of waves

$$\text{Power} = \frac{1}{f} \qquad (f \text{ is in } m)$$

(Power is in diopters)

c) $\text{Power} > -0.25\,D$

What is the min far point of the eye?



$$d_o = \infty \qquad d_i = -d_{far}$$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\frac{1}{f} = \frac{1}{\infty} - \frac{1}{d_{far}}$$

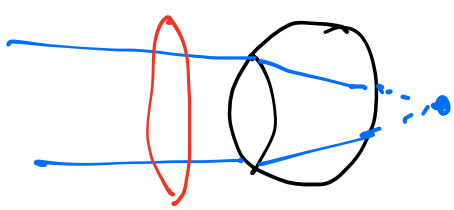
$$f = -d_{far}$$

$$P = \frac{1}{f} = -\frac{1}{d_{far}}$$

$$-\frac{1}{d_{far}} > -0.25$$

$$\frac{1}{d_{far}} < 0.25$$

$$4_m < d_{far}$$



Lens maker's formula

- General formula for determing focal length of non-thin lenses

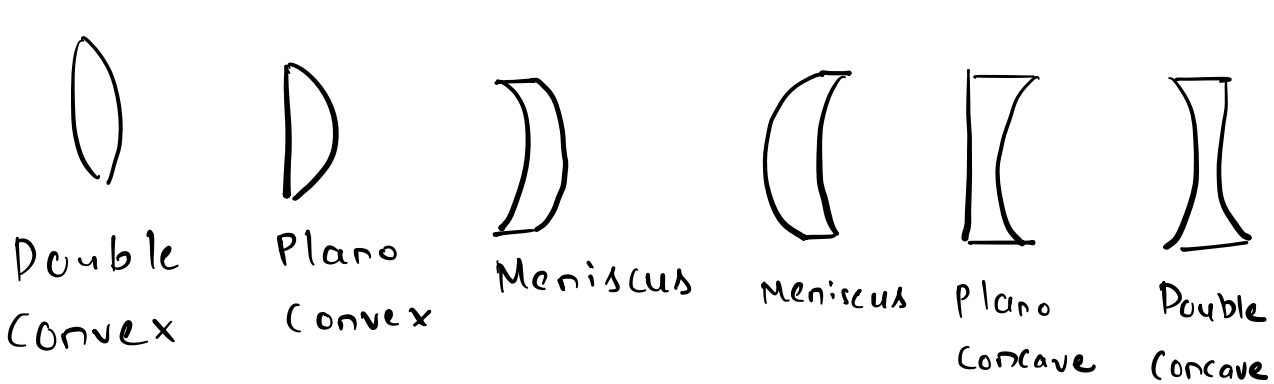
$$\frac{1}{f} = \left(\frac{n_{lens}}{n_{medium}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

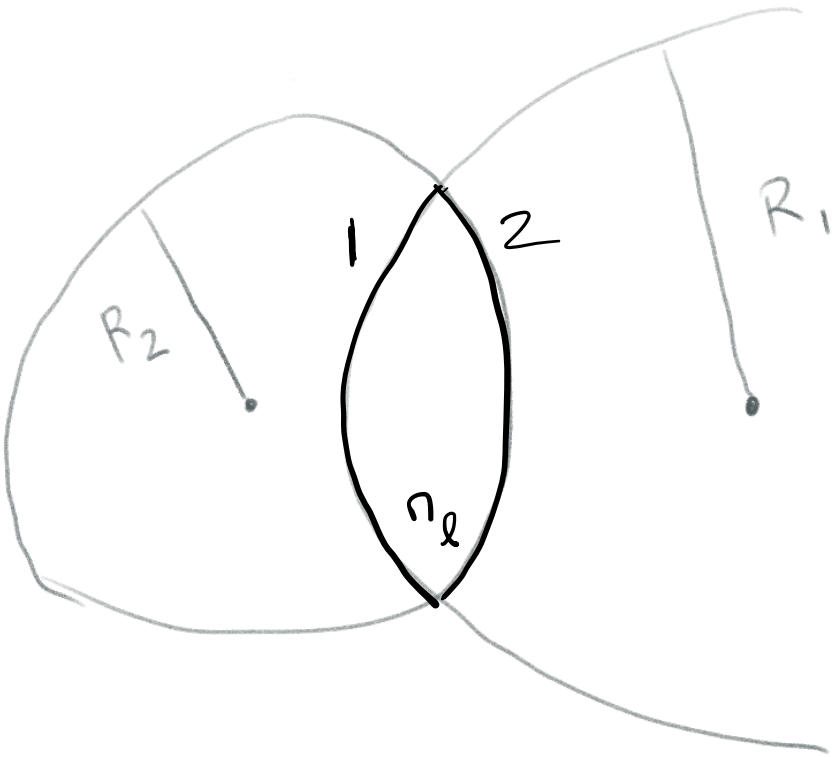
IMP

Indices of refraction

Radii of curvature

$$P = \frac{1}{f}$$





Center of sphere is on left: $R<0$

" right: $R>0$

Above example: $R_1>0$

$R_2<0$

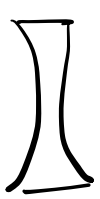
$f>0$: Converging lens

$f<0$: Diverging lens

Alternate heuristic

- If side towards object is convex $\Rightarrow R > 0$

(front of spoon = convex
back of spoon = concave)

$R_1 > 0$	$R_1 = \infty$	$R_1 < 0$	$R_1 > 0$	$R_1 = \infty$	$R_1 < 0$
$R_2 < 0$	$R_2 < 0$	$R_2 < 0$	$R_2 > 0$	$R_2 > 0$	$R_2 > 0$
$f > 0$	$f > 0$	$f = ?$	$f = ?$	$f < 0$	$f < 0$
					
Double Convex	Plano Convex	Meniscus	Meniscus	Plano Concave	Double Concave

- Lenses thinner in the middle are diverging
- Lenses thicker in the middle are converging

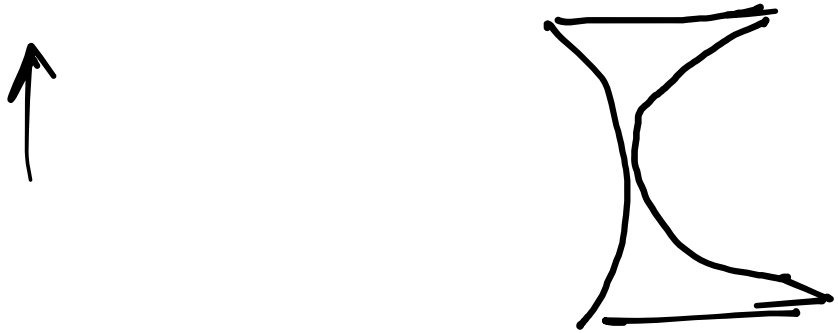
26.8 Lensmaker

The two surfaces of a lens have radii of $r_1 =$ - 20 *cm* and $r_2 =$ + 5 *cm*. The index of refraction of the lens is $n =$ 1.52. What kind lens is it?

A) Converging
B) Diverging
C) Not enough information

$$\frac{1}{f} = \left(\frac{n_{glass}}{n_{air}} - 1\right)\left(\frac{-1}{20} - \frac{1}{5}\right)$$

$$= 0.52\left(-\frac{5}{20}\right) \qquad f < 0$$



Interference of waves

- Waves are a disturbance in a medium that transmit information / energy through that medium

$$V = f\lambda$$

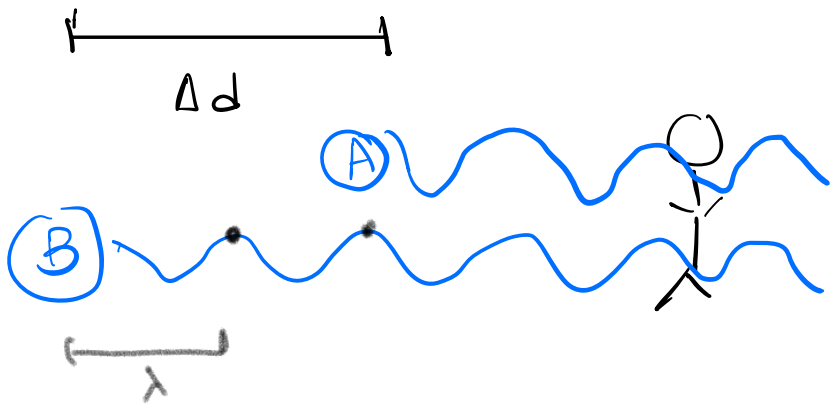
- V: wave speed determined by medium
- f: Frequency
- λ: Wavelength in medium

Constructive -: Waves overlap in phase
amplitudes add

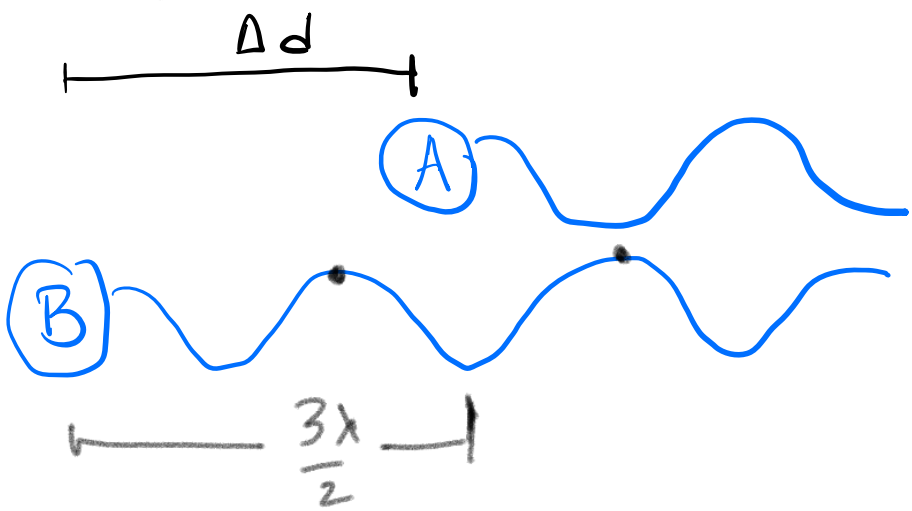
Destructive -: Waves overlap out of phase
amplitudes subtract

- Sources A & B emit waves in phase

Constructive interference



Destructive interference



$$\Delta d = \begin{cases} m\lambda & \text{Cons.} \\ \left(m+\frac{1}{2}\right)\lambda & \text{Dest.} \end{cases}$$

Demo link: https://phet.colorado.edu/sims/html/wave-on-a-string/latest/wave-on-a-string_all.html